

Guanlin Yu

Greater Seattle Area (Open to Relocation). US Citizen, no need for VISA sponsorship.

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SKILLS

Language: English(native), Chinese(native), Japanese(fluent), French(fluent)

Programming Language: Java, Python, Go, JavaScript, C#, C++

Tools & Frameworks: Spring/SpringBoot, Node.js, Django, Kubernetes, gRPC, Protocol Buffer, React, HTML, CSS, SQL(MySQL), NoSQL(Redis, Memcached), Message Queue(AWS SQS, RocketMQ, Kafka), Git, Agile

AWS: S3, SNS, SQS, EC2, ECS, DynamoDB, API Gateway, Lambda function, CloudWatch

Platform/OS: Linux, MacOS, Windows

INTERNSHIP EXPERIENCE

Reality AI (Software Developer Intern)

01/2025 – present (Remote)

- Partnered with cross-functional teams to design and implement AI-driven solutions aligned with business goals, leveraging Java and **TensorFlow**.
- Performed in-depth data analysis and preprocessing to deliver high-quality inputs for machine learning models, utilizing **Pandas** for data manipulation.
- Designed, implemented, and fine-tuned machine learning algorithms, achieving over a 25% improvement in accuracy using frameworks such as **PyTorch**.
- Deployed end-to-end AI systems, seamlessly integrating them with existing infrastructure, ensuring scalability through **Docker** containerization.

OSCR AI (Software Developer Intern)

LA, California, USA

Marketing Integration Project

10/2024 - 2/2025

- Built and integrated a **React**-based frontend component to enable affiliate marketing conversion tracking, improving business analytics capabilities.
- Enhanced marketing performance visibility by enabling tracking metrics for conversion rates, leading to a 20% improvement in affiliate sales conversion efficiency.
- Used React hooks for **state management** to ensure seamless data flow and React Router for navigation within the app.
- Conducted **API integration testing** using **Postman** and debugged issues using Chrome DevTools, reducing errors by 15%.
- Collaborated with cross-functional teams to align affiliate integration with business goals, completing the project 10% ahead of schedule.

Full-Stack AI-Powered Q&A Platform for PDF Document Queries

10/2024 - 12/2024

- Developed a full-stack React-based interactive web application using Ant Design, enabling seamless uploading and querying of PDF documents.
- Designed and implemented a robust backend with **Express.js** and **Multer** for file uploads and efficient API handling.
- Leveraged **LangChain** and **OpenAI GPT** APIs to load, split, and embed PDF content into a **memory vector** store for **real-time document-based query retrieval**.
- Featured both real-time voice interaction and speech/script interaction between user and web application.
- Integrated a Retrieval QA Chain for advanced **AI-driven responses**, ensuring high accuracy and scalability.
- Optimized the system for efficient processing of large documents and **natural language queries**.

SCEGC ENGINEERING DESIGN INSTITUTE (Web Application Developer Intern)

Shaanxi, China

Employee Timekeeping App Based on Facial Recognition Project

12/2023-02/2024(Remote)

- Utilized Streamlit to design and implement an easy-to-use user front-end interaction page.
- Core function utilized **MTCNN**, **InceptionResnetV1** model from facenet **Pytorch Machine Learning** library to obtain and compare the facial feature data.
- Solved the write-heavy backend need by integrating **PostgreSQL** driver psycopg to connect the database and update local and database data in both directions—improved timekeeping accuracy, decreasing timekeeping error follow-up rate by 30%.

SELECTED PROJECTS

Audio Media Android Project

09/2024-11/2024

- Designed a **Kotlin** Spotify-like favorite app by fully leveraging **Android Jetpack Library** and **Hilt Dependency Injection**.
- Implemented BottomBar and App Navigation using **Jetpack Navigation** components.
- Created Restful Api json-server and used **Retrofit** to handle requests.

- Built the feed/album/favorite **UI** components in **Jetpack Compose** following **MVVM** structure.
- Enabled local cache ability for favorite feature feed by using **Room** database.
- Integrated the **Google Exoplayer** to handle global music playback.

E-commerce Flash Sell Project

03/2024-07/2024

- Implemented a **distributed ID system (Snowflake ID algorithm)** to ensure unique order identifiers. Ensured unique order numbers while also enabling the query based on time stamps.
- Leveraged **Redis** for cache warming to reduce database load and improve response time.
- Ensured data consistency between payment and inventory reduction operations by first processing order inventory count in the cache. Incorporated **SpringBoot Segmented Locking** feature, separating order inventory deduction in different phases of ordering, improving system response during high traffic spikes.
- Utilized message middleware to smooth traffic spikes, integrating order processing with the **Message queue (Rocket MQ)** autoscaling feature. Stabilized database query traffic by Peak Loading Shifting using Lua script-enabled **asynchronous tasks**.
- Employed **Content Delivery Network (CDN)** for traffic distribution and static page rendering to alleviate server load. Bypassed dynamic page rendering, thus increasing performance by 60% under traffic spike.
- Integrated **Sentinel traffic control framework** with Spring Boot to manage and control traffic flows effectively.
- Utilized Jmeter to test traffic flow, ensuring system performance at 10,000 queries/second level.

Full-Stack SpringBoot Twitch Streaming Recommendation Web Application

11/2024-12/2024

A full-stack web application for users to search Twitch resources (stream/video/clip) and get recommendations.

- Built a web page with easy-to-use web components using React and Ant Design.
- Created RESTful APIs to handle HTTP requests and responses.
- Used MySQL database on Amazon Web Service RDS to store game data fetched from Twitch API.
- Supported OAuth2 login/logout and favorite collection.
- Deployed to Amazon Web Service App Runner for public access.

MINI SOCIAL MEDIA SERVICE

01/2023-09/2023

A full stack social media mobile app, featuring React.JS frontend and Python Django backend. Implemented accounts management, posts, static file storage and account relationship management features of modern social media application.

- Featured different models with proper permission checks and unit tests using the Django Test framework. Integrated CI/CD software development life cycle by ensuring all modules' unit tests pass before module deployment.
- Integrated RESTful APIs for dynamic content retrieval and implemented state management.
- Managed to improve data retrieval efficiency by integrating **Memcached** and **Redis** for friendship fanout, post pagination, and post management. Utilized **asynchronous tasks** to solve celebrity issues when fanout traffic is extremely large.
- Upgraded the post management and newsfeed fanout using **RateLimiter** to improve app response time—upgraded **relational database**, based on **HBase** to store friendships.
- Integrated **AWS S3** to store static files and user information to reduce server request pressure.

EDUCATION

M.S. in Computer Information System @ Bellevue University
Cumulative GPA: 3.92

12/2025(Expected)

B.S. in Pre Med @ University of Washington (Seattle, WA)

05/2021